

- DA 2013 -

DATA MANAGEMENT

Week 1: Introduction to Data Management

<>// ABOUT THIS COURSE – HOUSEKEEPING //>

Objectives:

- CLO1: demonstrate awareness of basic concepts in data management
- CLO2: apply suitable tools to merge data from variety of data sources and different data structures
- CLO3: identify and manage extreme observations and missing values in data
- CLO4: create new features aligning with the objectives of the analysis

No. of credits:

2 (10 weeks, 20 L – 20 P – 60 SS)

Assessment:

Quizzes- 10% (CLO1)

In-Class Assignments - 30% (CLO 2, 3)

Take Home Assignments - 20% (CLO 2, 3, 4)

Final Assessment - 40%



DA 2013 - Course
Plan

< // SOME GROUND RULES // >

- **Don't hesitate to ask questions** - no one knows 100% of anything
- **If your code works perfectly on the first try, check again...**
something's probably off!
- **It's okay to feel stuck** - understanding comes from trial, error, and curiosity.
- **Take a moment to think:** What do I need, and what am I trying to do?
- **Treat this like an adventure** - explore and have fun along the way!



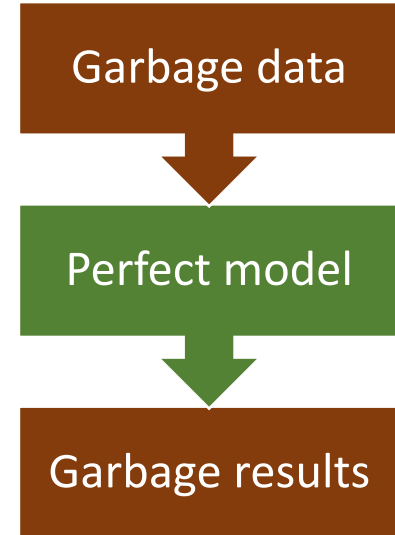
<>// TODAY'S OBJECTIVES //>

- The Data Lifecycle
- Types of Data
- Common Data Formats
- Tools for Handling Data
- Loading and Exploring Data
- Basic Data Quality



<>// DATA: THE WORLD'S BIGGEST ASSET //>

- Data drives decisions, innovation, and growth in every field.
- Every piece of data matters.
- Poor data = poor insights
- Even if your model is perfect, if your input data is not of good quality, the results will not be good.
This is called “**Garbage In, Garbage Out**” (GIGO).





LIFELINE OF A DATA ANALYST



**Data Analysts
spend 60% of
their time
cleaning data!**



Source: [Dataleum](#)



BEYOND DATA CLEANING



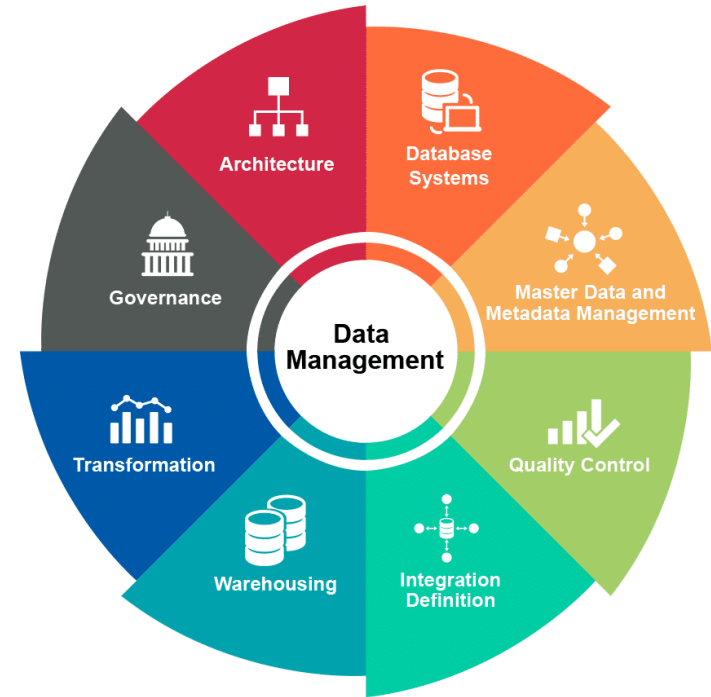
- Cleaning data improves its quality, but it's only one step in the journey to gaining the most out of data.
- Most of your data's value is probably lost in a multitude of databases and separate systems that aren't integrated through a data management strategy.
- Without proper storage, organization, and governance, even well-cleaned data quickly loses its value. On the other hand poor management makes cleaning complicated and more expensive than it needs to be.
- The key to success: **Effective data management**



WHAT IS DATA MANAGEMENT



- Data management is the **practice of collecting, organizing, protecting, and storing an organization's data so it can be analyzed for business decisions.**
- Managing data is as critical as safeguarding a bank - data is an asset, and lost data is lost wealth!



<>// COMPONENTS OF DATA MANAGEMENT //>

1. Data Acquisition

Collect data from internal and external sources with a focus on accuracy and relevance.

Internal sources

- Transactional systems
- CRM (Customer Relationship Management)
- ERP (Enterprise Resource Planning)
- Sensor/IoT feeds

External sources

- Open data portals (Kaggle, GitHub)
- APIs
- Surveys
- Web scraping
- Partner datasets (from third-parties)

<>// COMPONENTS OF DATA MANAGEMENT //>

2. Data Storage

Choose fit-for-purpose repositories (DBs, data warehouses, data lakes, cloud)

Plan for scalability and disaster recovery

Use efficient file formats and indexing for performance – keeping in mind the type of data that is involved in the project





COMMON DATA FORMATS



- CSV (.csv) – simple, widely supported, plain text.
- Excel (.xlsx) – multiple sheets, formulas, for business reporting.
- JSON(.json) – nested, human-readable (APIs, web data).
- XML(.xml) – tag-based and customizable, great for detailing structured layouts.
- Databases (.sql/ .db) – powerful querying, indexing, long-term storage.
- APIs (request/response) – live data delivery, dynamic updates, no file download.





DATA STORAGE: TYPES OF DATA



The way you manage each type of data differs. There are three main types of data:

1. **Structured data:** This includes data that is highly organized and stored in tabular formats, such as databases or spreadsheets.

Ex: A table of product prices in an online store's backend database.

2. **Unstructured data:** This includes data without a predefined format, such as plain text, images, videos, and audio files.

Ex: A blog article, YouTube video, or customer review.

3. **Semi-structured data:** This type of data has some organizational properties but does not reside in a strict tabular format. It is commonly found in formats like JSON, XML, or CSV.

Ex: API responses from weather data services or tweets in JSON format.



DIFFERENT DATA STORAGES



	 Database	VS	 Data Warehouse	VS	 Data mart	VS	 Data lake
Scope	Application-specific		Organization-wide, structured data.		Department-specific, structured data.		Organization-wide, any type of data
Data Type	Structured		Structured		Structured		Structured, semi-structured, unstructured.
Structure	Predefined schema		Schema on write		Schema on write (inherited from data warehouse)		Schema on read
Use Case	Operational applications(OLTP)		Business intelligence, historical analysis(OLAP).		Specific business function analysis		Big data analytics, data exploration.

Source: [differences-between-database-vs-data-warehouse-mart-lake](#)

<>// COMPONENTS OF DATA MANAGEMENT //>

3. Data Security & Privacy

Implement security measures and access controls to prevent unauthorized access, breaches, or exposure.

Security measures:

- Role-based access controls and encryption
- Masking/anonymisation of sensitive fields
- Compliance with regulations
- Regular security audits and penetration testing

<>// COMPONENTS OF DATA MANAGEMENT //>

4. Data Quality

Ensure accuracy and consistency by detecting and correcting errors, redundancies, and inconsistencies.

Steps to ensure data quality:

- Missing values: Drop, impute (median/mean), or flag.
- Duplicates: Remove exact duplicates or deduplicate by key columns.
- Incorrect types: Convert strings to numbers or dates (astype, to_datetime).
- Outliers: Check domain knowledge before removing.



DIMENSIONS OF DATA QUALITY



Accuracy

The degree to which the data correctly represents the entity or attribute being described.



Completeness

The percentage of missing data from a given data set.



Consistency

The absence of difference or contradiction in data irrespective of the data's source.



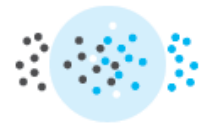
Validity

Invalid data affects the accuracy and completeness of a data set.



Integrity

The validity of relationships across various data entities.



Uniqueness

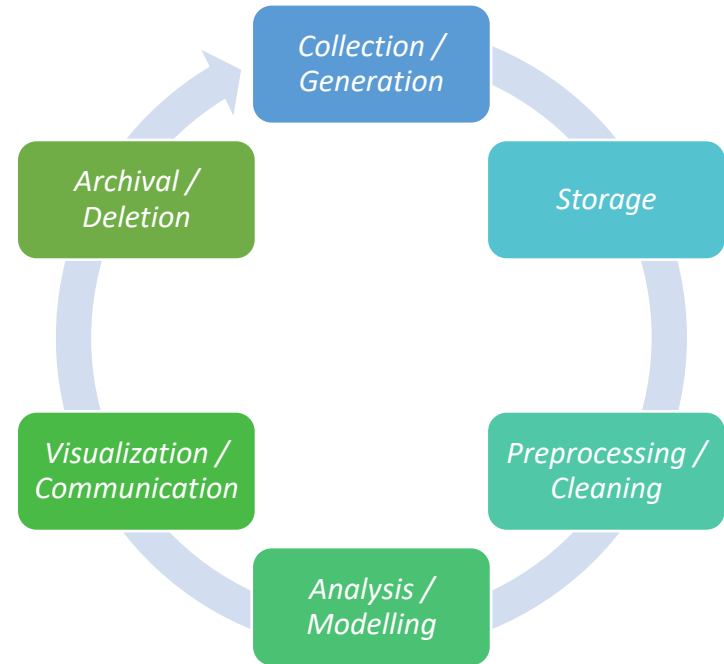
Ensures duplicate or overlapping data is identified and marked.

<>// COMPONENTS OF DATA MANAGEMENT //>

5. Data Lifecycle Management

Manage data from creation to disposal, meeting legal and regulatory requirements at every stage.

Data rarely stays in one place or form – it flows through stages. It is important to maintain artefacts to ensure there is evidence for proper lifecycle management.



<>// COMPONENTS OF DATA MANAGEMENT //>

6. Data Governance

Establish policies, standards, and ownership responsibilities to guide data use throughout its lifecycle

- Establish a governance board or data stewards
- Create policies, standards and naming conventions
- Define ownership and accountability for each dataset

<>// COMPONENTS OF DATA MANAGEMENT //>

7. Data Integration & Pipelines

Combine data from multiple sources to create a unified view for analytics and reporting

- Automate ingestion from multiple sources (ETL / TPA Extracts)
- Apply transformations and data enrichment
- Maintain change logs

<>// COMPONENTS OF DATA MANAGEMENT //>

8. Data Analytics & Insight Generation

Analyze data to extract insights, support decision-making, and uncover trends and patterns

- Exploratory analysis and dashboards
- Advanced modelling: statistics, machine learning, forecasting
- Translate insights into business actions and KPIs

<>// COMPONENTS OF DATA MANAGEMENT //>

9. Data Documentation & Metadata Management

Maintain clear documentation and metadata to give context and improve understanding of data assets

- Maintain data dictionaries, metadata repositories
- Document business rules and transformations
- Make data discoverable with catalogues and search tools

<>// COMPONENTS OF DATA MANAGEMENT //>

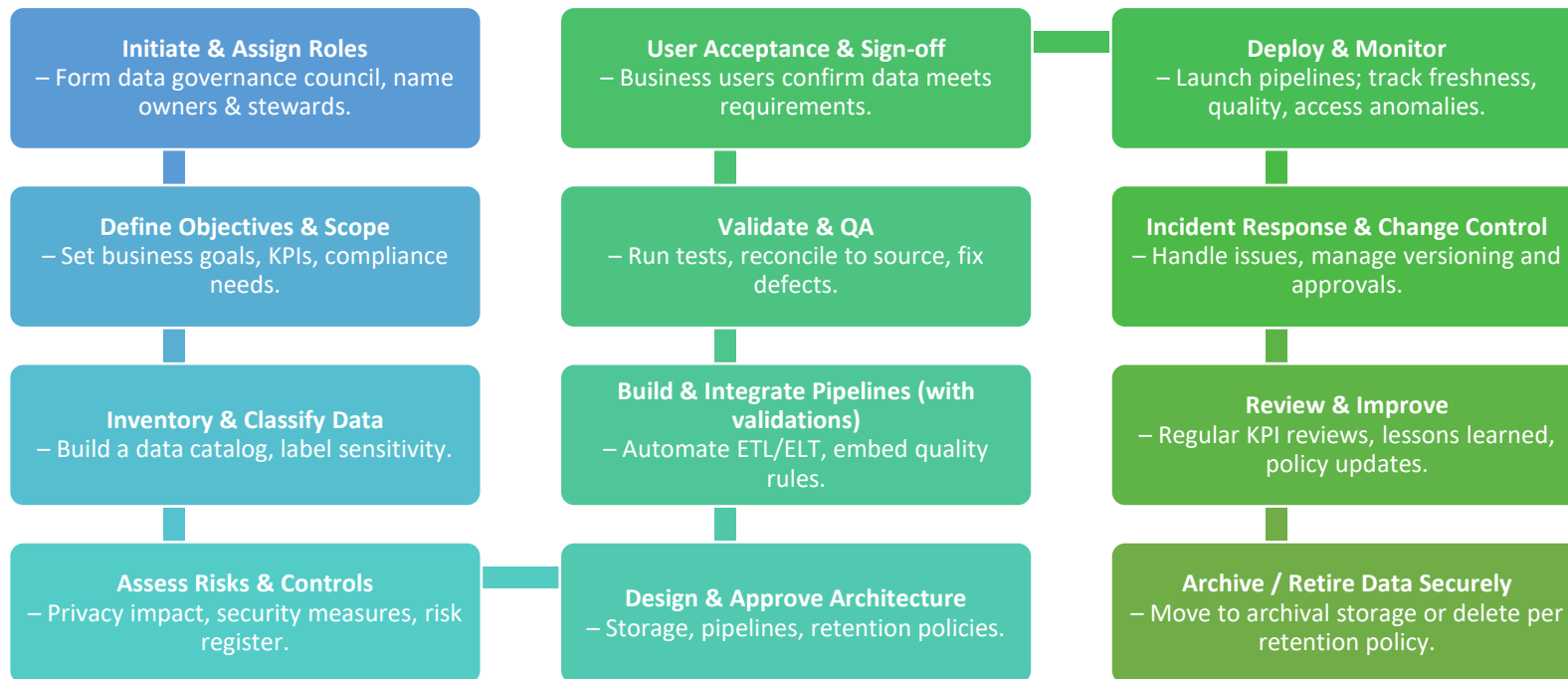
10. Ongoing Monitoring & Version Control

Track data lineage, monitor system performance, maintain historical versions and change logs, and use feedback loops to improve every stage of the cycle over time.

- Continuous monitoring of data quality, security, and usage
- Version control for datasets, schemas, and analytics models
- Feedback loop to update acquisition, governance, or storage strategies



DATA GOVERNANCE IN PRACTICE





TOOLS FOR HANDLING DATA



- **Python:** Flexible, code-based cleaning & analysis.
- **Excel:** Quick exploration, pivot tables.
- **SQL:** Efficient querying of large datasets.
- **R:** Statistical modelling and plotting.
- **PowerBI / Tableau:** Data visualization and dashboard generation.





SOURCES FOR DATA PROJECTS



1. Open Data Platforms

- [Kaggle Datasets](#) – wide variety of datasets for analytics and ML projects
- [UCI Machine Learning Repository](#) – classic ML datasets
- [data.gov](#) – US government open data

2. Free APIs for Live Data

- [OpenWeatherMap API](#) – weather data
- [Twitter API](#) – social media data
- [JSONPlaceholder](#) – sample JSON API for practice

3. Public Research Datasets

- [Google Dataset Search](#) – search across academic/public datasets
- [World Bank Data](#) – economic, development, and social datasets



SOURCES FOR DATA PROJECTS



4. Locally Stored data & Survey Data

- CSV, Excel, SQL databases collected internally
- Surveys, forms, interviews, or organizational logs
- Ideal for projects tailored to specific business or research questions

5. Company / Open-Source Projects

- GitHub repositories with sample data
- Public competitions and hackathons

Special notes:

- Always check **licenses, privacy, and usage restrictions** before using datasets.
- Medical data / personal identifiable data have to be handled carefully.

<>// DATA MANAGEMENT BEST PRACTICES //>

Clearly Identify Business Goals

- Start by defining your organization's objectives.
- Avoid storing irrelevant data and keep systems manageable.
- Ensure that the tools and methods chosen are conceptually sound and fit-for-purpose.

Focus on Data Quality

- Reliable analytics depend on high-quality data.
- Streamline collection and storage and perform regular accuracy checks.
- Detect and correct inconsistent formats, spelling errors, and outdated information.

Ensure the Right Access

- Quality data must be usable by the right people.
- Use tiered permissions instead of blanket access to balance convenience and security.
- Efficient access prevents wasted time and resources.

Prioritize Data Security

- Protect data from external threats and comply with regulations.
- Train staff on correct handling procedures.
- Have a breach-response plan and choose software that supports security best practices.



TO DO LIST BEFORE WEEK 2



- Install Python 3.x
- Create a Kaggle account.
- Read more about the importance of data management and relevant job roles in the market
- Go to the [Pandas](#) web page and be familiar with the functions offered.



THANK YOU

<SEE YOU NEXT WEEK!>

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